

EXPERIMENT 5:

AIM:

To create a bar chart representing the trading volume of Alphabet Inc. stock between two dates.

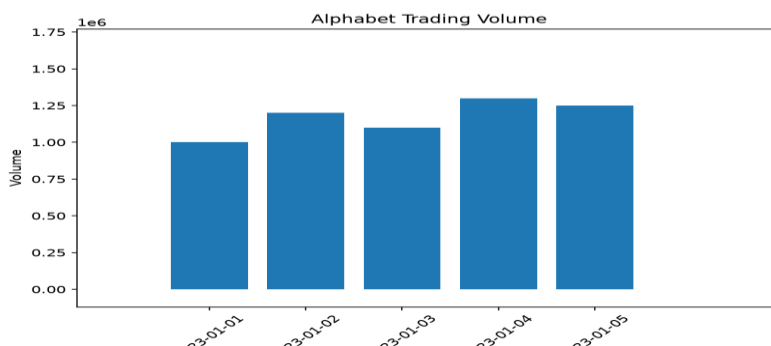
INPUT:

Date	Close	Volume
2023-01-01	95	1000000
2023-01-02	97	1200000
2023-01-03	99	1100000
2023-01-04	101	1300000
2023-01-05	100	1250000

PROGRAM:

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 df = pd.read_excel(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp 4,5,6_data.xlsx")
4 df["Date"] = pd.to_datetime(df["Date"])
5 data = df[(df["Date"] >= "2023-01-01") &
6           (df["Date"] <= "2023-01-05")]
7 plt.figure(figsize=(8,5))
8 plt.bar(data["Date"].dt.strftime("%Y-%m-%d"), data["Volume"])
9 plt.title("Alphabet Trading Volume")
10 plt.xlabel("Date")
11 plt.ylabel("Volume")
12 plt.xticks(rotation=45)
13 plt.show()
```

OUTPUT:



RESULT:

The trading volume of **Alphabet Inc.** stock between the specified dates was successfully represented using a **bar chart**, clearly showing the daily trading volumes.

EXPERIMENT 6:

AIM:

To create a scatter plot showing the relationship between trading volume and closing stock prices.

INPUT:

Date	Close	Volume
2023-01-01	95	1000000
2023-01-02	97	1200000
2023-01-03	99	1100000
2023-01-04	101	1300000
2023-01-05	100	1250000

PROGRAM:

```
exp6.py > ...
1  import pandas as pd
2  import matplotlib.pyplot as plt
3  df = pd.read_excel(r"C:\Users\dhars\Documents\Query processing\lab_questions\exp 4,5,6_data.xlsx")
4  df["Date"] = pd.to_datetime(df["Date"])
5  data = df[(df["Date"] >= "2023-01-01") &
6           (df["Date"] <= "2023-01-05")]
7  plt.figure(figsize=(8,5))
8  plt.scatter(data["Volume"], data["Close"])
9  plt.title("Trading Volume vs Closing Price")
10 plt.xlabel("Trading Volume")
11 plt.ylabel("Closing Price")
12 plt.grid(True)
13 plt.show()
```

OUTPUT:



RESULT:

A **scatter plot** was successfully created to visualize the relationship between the **trading volume** and the **closing stock prices** of Alphabet Inc. stock between the specified dates.